5.2 Basis & Dimension computations

* calculate bases for subspaces
* transpose switches columns/rows
* $Col\ A = Row\ A^T$
  $Col\ A^T = Row\ A$
* $V = Col A \Rightarrow V^\perp = (Col A)^\perp = (Row A^T)^\perp = Null\ A^T$
  $Row^\perp = Null$
  ↳ $RA^T = Nul$

① Bases for Col A & Row A

- Col A basis = pivot columns of A
- Row A basis = non-zero rows of A_{red}

⇒ Rank A = dim Col A = dim Row A = # pivots in A_{red}

② Basis for Null A

- $[A \mid 0] \rightarrow [A_{red} \mid 0] \rightarrow \underline{x} = t_1\underline{x}_1 + \dots + t_k\underline{x}_k$
- free parameters (per free column in A_{red}) → basis $\{\underline{x}_1, \dots, \underline{x}_k\}$

⇒ dim Null A = # of free columns

③ Rank Theorem

- $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $A - m \times n$
- dim Col A + dim Null A = dim $\mathbb{R}^n$ = n } Col A & Null A complementary
- # pivot col + # free col = n

④ Basis for $V^\perp$

- $V^\perp = (Col A)^\perp = (Row A^T)^\perp = Null\ A^T$

⇒ take transpose & find basis for Null A^T

⑤ Basis for Col A & Col $A^\perp$

- Col A = Row A^T
  Col $A^\perp$ = Null A^T
  ⇒ $A \rightarrow A^T \rightarrow A^T_{red}$ ↗ Col A = pivot rows as columns
  ↘ $[A^T_{red} \mid 0]$ = Col$A^\perp$ basis
- dim V + dim $V^\perp$ = dim $\mathbb{R}^n$ = n (V = Col A)

⑥ Basis for $\mathbb{R}^n$

- If $V \cap V^\perp = \{\underline{0}\}$ and dim V + dim $V^\perp$ = n ⇒ union is basis for $\mathbb{R}^n$
- To extend basis:
  ↙ $NA^T = ColA^\perp$
  $A \rightarrow A^T \rightarrow$ basis for Null A^T = Col $A^\perp$ → Union w/ Col A

6.1, 6.2 Matrix Algebra

① Matrices & Linear Transformations

* sum of transformations
  - $(T_A + T_B)(\underline{x}) = T_A(\underline{x}) + T_B(\underline{x})$
  - $T_A + T_B = T_{A+B}$
  - operation of transformation = operation of matrices
* scale of transformations
  - $(\lambda T_A)(\underline{x}) = \lambda(T_A(\underline{x}))$
  - $\lambda T_A = T_{\lambda A}$
  - scaling transformation = scaling outputs

② Matrix Multiplication

* $[\quad][\quad] = [\cdot]$ w/ dotless product
* Facts
  - AB ≠ BA
  - A (B+C) = AB + BC
  - (A+B)C = AC + BC
  - $A(\lambda B) = \lambda(AB) = (\lambda A)B$
  - A(BC) = (AB)C
* matrix multiplication = composition of linear transformations
  - $I_m \xrightarrow{\text{row op}} E$ ⋯→ $A \rightarrow EA$ (multiplication on <u>left</u> by elementary matrix performs the row operations) = $[I_n]$

③ Matrix Transpose w/ operations

* facts:
  - $(A+B)^T = A^T + B^T$
  - $(\lambda A)^T = \lambda(A^T)$
  - $(AB)^T = B^T A^T$ (order reversed)
* consequence:
  - $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ ($\underline{v}$)
    $T_A^T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ ($\underline{u}$)
    ⇒ $\underline{u} \cdot (A\underline{v}) = (A^T\underline{u}) \cdot \underline{v}$
    ↳ $\underline{u} \cdot (A\underline{v}) = \underline{u}^T(A\underline{v}) = (\underline{u}^T A)\underline{v} = (A^T\underline{u})^T\underline{v} = (A^T\underline{u})^T\underline{v}$

7.1 Invertible Matrices

① Invertibility

- $n \times n$ matrix A, $n \times n$ matrix B invertible if
  $AB = I_n = BA$, $B = A^{-1}$
- $AB = I_n \Leftrightarrow BA = I_n$
- $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ invertible $\Leftrightarrow$ $ad - bc \neq 0$ & $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
- elementary matrices are invertible w/ elementary matrices
- Properties
  - $(A^{-1})^{-1} = A$
  - $(AB)^{-1} = B^{-1}A^{-1}$
  - $I_n^{-1} = I_n$
  - $(A^T)^{-1} = (A^{-1})^T$
  - $(cA)^{-1} = c^{-1}A^{-1}$
  - A, B invertible $\not\Rightarrow$ $A+B$ invertible

② usage

- A invertible & $A\underline{x} = \underline{b}$ $\Rightarrow$ $\underline{x} = A^{-1}\underline{b}$
- A invertible $\Rightarrow$ $A\underline{x} = b$ has unique solution for all $b \in \mathbb{R}^n$ $\Rightarrow$ $A_{red} = I_n$

* A invertible $\Leftrightarrow$ $A_{red} = I_n$
* $[A \mid I_n]_{red} = [I_n \mid A^{-1}]$ (b/c A invertible $\Leftrightarrow$ product of elementary matrices)

③ Conceptual

- $AB = I_n = BA \Leftrightarrow T_A(T_B(\underline{x})) = \underline{x} = T_B(T_A(\underline{x}))$
- A invertible w/ inverse A^{-1} $\Leftrightarrow$ T_A invertible w/ inverse $T_{A^{-1}}$

④ Goldilocks Principle

7.2, 7.3 Triangular & Orthogonal Matrices

① Triangular Matrices

* lower triangular = zero above main diagonal
* upper triangular = zero below main diagonal
* diagonal matrices = upper & lower
* properties
  - product of 2 upper triangular = upper triangular
  - A $n \times n$ $\Rightarrow$ A_{red} upper triangular
  - row equivalent to I_n: $\begin{bmatrix} a_1 & & * \\ & \ddots & \\ 0 & & a_n \end{bmatrix}$ invertible $\Leftrightarrow$ $a_1 \neq 0, \ldots, a_n \neq 0$
  - A invertible $\Leftrightarrow$ columns of A form basis for $\mathbb{R}^n$

② Orthogonal Matrices

- $\{\underline{v}_1, \ldots, \underline{v}_n\} \subset \mathbb{R}^n$ orthogonal basis if:
  - $\underline{v}_i \cdot \underline{v}_j = 0$ (pairwise orthogonal)
- orthonormal basis if:
  - $\underline{v}_i \cdot \underline{v}_j = 0$
  - $\underline{v}_i \cdot \underline{v}_i = 1$ for all i
- Facts
  - non-zero pairwise orthogonal vectors are L.I. (form basis)
  - $U = [\underline{v}_1 \ldots \underline{v}_n]$ orthogonal $\Leftrightarrow$ $\{\underline{v}_1, \ldots, \underline{v}_n\}$ orthonormal basis
  - U a 2×2 matrix $\Leftrightarrow$ U a 2×2 rotation/reflection matrix
- General properties

U orthogonal

$U^T U = I_n = U U^T$ $\Leftrightarrow$ U orthogonal $\Leftrightarrow$ $\|U\underline{x}\| = \|\underline{x}\|$ (preserve length)

$U^T U = I_n = U U^T$ $\Leftrightarrow$ $(U\underline{x}) \cdot (U\underline{y}) = \underline{x} \cdot \underline{y}$ (preserve dot product) $\Leftrightarrow$ $\|U\underline{x}\| = \|\underline{x}\|$

* orthogonal operations only change direction, not length *

8.1, 8.2 Determinants

① Determinant Basics

- $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ invertible $\Leftrightarrow$ $ad - bc \neq 0$
- det = 0 $\Leftrightarrow$ A not invertible
- det A ≠ 0 $\Leftrightarrow$ Area (R) ≠ 0
  $\Leftrightarrow$ columns of A a basis for $\mathbb{R}^2$
  $\Leftrightarrow$ A invertible
- Area (R) = |det A|
- $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ scales areas by |det A|
- signs
  ↳ det A > 0 : columns oriented in same way standard basis
  ↳ det A < 0 : columns oriented different way to standard basis
  ↳ det A = 0 : ⟶ lose dim

- det (A) + det (B) ≠ det (A+B)
- determinants not linear

② Calculating Determinant

- Permutation Matrix
  - P is a permutation matrix if columns are the standard basis vectors in any order
  - $\text{sign}(P) = \begin{cases} +1 & \text{even \# of pairs out of order} \\ -1 & \text{odd \# of pairs out of order} \end{cases}$
- Elementary matrices
  - $A \xrightarrow{Ex} A'$ w/ $\det A' = -\det A$
    $A \xrightarrow{Sc} A'$ w/ $\det A' = c \det A$
    $A \xrightarrow{Sh} A'$ w/ $\det A' = \det A$
  - multiply row ops together until upper triangular
  - $\det AB = \det A \det B$
  - A invertible $\Leftrightarrow$ $\det A \neq 0$ & $\det A^{-1} = 1/\det A$
  - U orthogonal ⇒ det U = 1 (for rotations) or det U = -1 (reflections in $\mathbb{R}^2$)
- Cofactor Expansion
  - $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \to$

9.1, 9.2 Linear Coordinate Systems

① Basics

- $\underline{v} \in V \subset \mathbb{R}^n$ subspace
  $B = \{\underline{v}_1, \dots, \underline{v}_n\} \subset V$ basis
  ⇒ $\underline{v} = a_1\underline{v}_1 + \dots + a_d\underline{v}_d$ & $a_1, \dots, a_d$ unique for $\underline{v}$
- fixing B gives a grid
- $\underline{v} \in V \subset \mathbb{R}^n$ subspace
  $B = \{\underline{v}_1, \dots, \underline{v}_n\} \subset V$ basis
  $\underline{v} = a_1 v_1 + \dots + a_n v_n$
  ⇒ $[\underline{v}]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix} \in \mathbb{R}^d$ ← weights stacked in a column
  ↑ "B-coordinates of $\underline{v}$"
- B-coordinate system on V = a linear coordinate system
- * $P_B = [\underline{v}_1 \dots \underline{v}_d]$
  $P_B [\underline{v}]_B = \underline{v}$ ← "B-coords back to regular coords w/ matrix mul"
- $\mathcal{E}_n = \{\underline{e}_1, \dots, \underline{e}_n\}$ standard basis for $\mathbb{R}^n$
  - $V = \mathbb{R}^n$
  - $[\underline{v}]_{\mathcal{E}_n} = \underline{v}$ ← $\underline{v} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = a_1\underline{e}_1 + \dots + a_n\underline{e}_n$
    ↑ standard coordinates

② Calculating Coordinates (B-coordinates)

- $\underline{v} \in V \subset \mathbb{R}^n$
  $B = \{\underline{v}_1, \dots, \underline{v}_d\} \subset V$ basis
  ⇒ $[\underline{v}_1 \dots \underline{v}_d \mid \underline{v}]_{red} = \left[\begin{array}{c|c} I_d & [\underline{v}]_B \\ 0 & 0 \end{array}\right]$
  $\quad = [P_B \mid \underline{v}]$
- ⇒ $[\underline{v}_1]_B = e_1, \dots, [\underline{v}_d]_B = e_d$
  ↳ B as standard basis in B-coords
- $[c_1\underline{w}_1 + \dots + c_k w_k]_B = c_1[\underline{w}_1]_B + \dots + c_k[\underline{w}_k]_d$ → B-coords are linear

③ Change of Basis Matrix

- $V \subset \mathbb{R}^n$ subspace
  $B = \{\underline{b}_1, \dots, \underline{b}_d\} \subset V$
  $C = \{\underline{c}_1, \dots, \underline{c}_d\} \subset V$ bases
  ⇒ $\underset{C \leftarrow B}{P} = [[\underline{b}_1]_C \dots [\underline{b}_d]_C]$
  ↑ basis B in C-coords
- * $[\underline{c}_1 \dots \underline{c}_d \mid \underline{b}_1 \dots \underline{b}_d]_{red} = [P_C \mid P_B]_{red} = \left[\begin{array}{c|c} I_d & \underset{C \leftarrow B}{P} \\ 0 & 0 \end{array}\right]$
- $\underset{C \leftarrow B}{P} [\underline{v}]_B = [\underline{v}]_C$ ← translate btwn coord systems

$$\begin{tikzcd} & V \arrow[dl] \arrow[dr] & \\ \mathbb{R}^d \arrow[rr, "\underset{C \leftarrow B}{P}"] & & \mathbb{R}^d \end{tikzcd}$$

$[\underline{v}]_B \qquad [\underline{v}]_C$

B → standard → C

- $V = \mathbb{R}^n$ ⇒ $\underset{\mathcal{E}_n \leftarrow B}{P} = P_B$ & $\underset{C \leftarrow \mathcal{E}_n}{P} = P_C^{-1}$ ⇒ $\underset{C \leftarrow B}{P} = P_C^{-1} P_B$

10.1, 10.2 Orthogonal Bases and Coordinates

① Recall

- $v \in V \subset \mathbb{R}^n$ subspace

  $B = \{v_1, \dots, v_d\} \subset V$ basis $\Rightarrow [v]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^d$

  $v = a_1 v_1 + \dots + a_n v_n$

  ↑ B-coords for v

  $\Rightarrow \underbrace{[P_B \mid v]}_{[v_1 \dots v_d]}{}_{red} = \left[\begin{array}{c|c} I_n & [v]_B \\ 0 & 0 \end{array}\right]$

② Orthogonal / orthonormal change of Basis

- can skip G.E
- $B = \{v_1, \dots, v_d\} \subset V$ orthogonal basis $\Rightarrow [v]_B = \begin{bmatrix} \frac{v \cdot v_1}{v_1 \cdot v_1} \\ \vdots \\ \frac{v \cdot v_d}{v_d \cdot v_d} \end{bmatrix} \in \mathbb{R}^d$
- $B = \{v_1, \dots, v_d\} \subset V$ orthonormal basis $\Rightarrow [v]_B = \begin{bmatrix} v \cdot v_1 \\ \vdots \\ v \cdot v_d \end{bmatrix} = \begin{bmatrix} v_1^T v \\ \vdots \\ v_d^T v \end{bmatrix} = P_B^T v$
- B & C bases and C orthonormal $\Rightarrow P_{C \leftarrow B} = P_C^T P_B$

  ↳ orthog matrix if B & C orthonormal

③ Construct Orthonormal Basis

- $\underset{\text{basis for } V}{B = \{v_1, \dots, v_d\}} \xrightarrow{\text{G.S.}} \underset{\text{orthogonal basis } V}{U = \{u_1, \dots, u_d\}} \xrightarrow{\text{normalize}} \underset{\text{orthonormal basis}}{\hat{U} = \{\hat{u}_1, \dots, \hat{u}_d\}}$
- Gram-Schmidt Process
  - given $\{v_1, \dots, v_d\} \subset V \longrightarrow$ orthogonal basis
  - $u_1 = v_1$
  - $u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1$
- normalize
  - $\hat{u}_n = \frac{u_n}{\|u_n\|} = \frac{1}{\sqrt{u_n \cdot u_n}} u_n$

11.1 Projection Transformation

① Definition

- $V, W \subset \mathbb{R}^n$ subspaces are complements written $V \oplus W$ $\Leftrightarrow$ every $x \in \mathbb{R}^n$ can be written uniquely as $x = \underset{\text{in } V}{x_v} + \underset{\text{in } W}{x_w}$ $\Leftrightarrow$ $\dim V + \dim W = n$ & $V \cap W = \{0\}$
- Notes
  - can project at different angles
  - W = how we project
  - V = what we projecting to
  - $V \subset \mathbb{R}^n$ can have infinitely many complements
- $V, W \subset \mathbb{R}^n$ complements

$$\begin{tikzcd} & \underset{x}{\mathbb{R}^n} \arrow[dl, "P_{V\oplus W}"'] \arrow[dr, "P_{W\oplus V}"] & \\ \underset{x_v}{\mathbb{R}^n} & & \underset{x_w}{\mathbb{R}^n} \end{tikzcd}$$

* $P_{V \oplus W}(x) + P_{W \oplus V}(x) = x$ one proj gives another

$\Rightarrow P_{W \oplus V} = x - P_{V \oplus W}(x)$

- Algorithm
  - $P_{V \oplus W}(x) = x_v$ & $P_{W \oplus V}(x) = x_w = x - x_v$
  - Q: Find a basis for $V\ \{v_1, \dots, v_k\}$ & $W\ \{w_1, \dots, w_k\}$.

    B/C V, W comp: $\{v_1, \dots, v_k, w_1, \dots, w_k\}$ basis for $\mathbb{R}^n$,

    $x = \underbrace{a_1 v_1 + \dots + a_d v_d}_{x_v} + \underbrace{b_1 w_1 + \dots + b_k w_k}_{x_w}$

    $\Rightarrow$ solve: $[v_1 \cdots v_d\ w_1 \cdots w_k \mid x]$

② Special Case – Orthogonal Projection

avoids G.E.

- $V \subset \mathbb{R}^n$ subspace, $V^\perp = \{x \mid x \cdot v = 0\ \forall v \in V\}$
- $W = V^\perp$, $P_{V \oplus V^\perp} = P_V$
- $B = \{v_1, \dots, v_d\} \subset V$ orthogonal basis (↑ G.S.) $\Rightarrow$ $P_V(x) = \frac{x \cdot v_1}{v_1 \cdot v_1} v_1 + \dots + \frac{x \cdot v_d}{v_d \cdot v_d} v_d$

  $P_{V^\perp}(x) = x - P_V(x)$
- orthonormal basis $\Rightarrow$ $P_V(x) = P_B P_B^T (x)$ ← $P_B = [v_1 \dots v_d]$

  ↳ coef. matrix for projection
- if V = line & $V^\perp$ = hyperplane, project x to $V^\perp$ & $P_V(x) = x - P_{V^\perp}(x)$

③ Proj. Abstract Characterization

- $T: \mathbb{R}^n \to \mathbb{R}^n$ is a projection $\Leftrightarrow A^2 = A$

  $T = P_{V \oplus W}$ ↑ ↳ orthogonal projection if $A^T = A$

11.2 Geometry of Orthogonal Projections

① Orthogonal Projections

- $\underline{x} \in V \subset \mathbb{R}^n$ a subspace, $\underline{x} = \underline{x}_V + \underline{x}_{V^\perp}$
- $B = \{\underline{v}_1, \dots, \underline{v}_d\} \subset V$ orthogonal basis ⇒ $P_V(\underline{x}) = \frac{\underline{x}\cdot\underline{v}_1}{\underline{v}_1\cdot\underline{v}_1}\underline{v}_1 + \dots + \frac{\underline{x}\cdot\underline{v}_d}{\underline{v}_d\cdot\underline{v}_d}\underline{v}_d$

  G.S.

  $P_{V^\perp}(\underline{x}) = \underline{x} - P_V(\underline{x})$

② Minimum Distance

- minimum distance btwn $\underline{x}$ and V:

  $= \|\underline{x} - P_V(\underline{x})\| = \|P_{V^\perp}(\underline{x})\|$ ← $\le \|\underline{x} - \underline{y}\|$, $\underline{y}$ = closest

③ Least-Squares

- If $A\underline{x} = \underline{b}$ inconsistent ($\underline{b} \notin \mathrm{Col}\,A$), best approximate solution
- least-squares solution to $A\underline{x} = \underline{b}$ = solution to $A\underline{x} = P_{\mathrm{Col}\,A}(\underline{b})$ = solution to $A^T A\underline{x} = A^T\underline{b}$
- If $A\underline{x} = \underline{b}$ consistent ⇒ $\underline{b} = P_{\mathrm{Col}\,A}(\underline{b})$ ⇒ Least squares sol = ordinary sol.
- Method 1: find $P_{\mathrm{Col}\,A}(\underline{b})$ & solve $A\underline{x} = P_{\mathrm{Col}\,A}(\underline{b})$

  ↳ given A, do G.S. ↳ G.E SLE

  $\|P_{\mathrm{Col}\,A}(\underline{b}) - \underline{b}\|$ = length

④ Normal Equations

- Fact: $A\underline{x} = P_{\mathrm{Col}\,A}(\underline{b}) \Leftrightarrow A^T A\underline{x} = A^T\underline{b}$

  ↑ normal Eq. of $A\underline{x} = \underline{b}$ & solve w/o $P_{\mathrm{Col}\,A}(\underline{b})$
- Method 2: $[A^T A \mid A^T\underline{b}]$

  ↑ $\underline{b} - A\underline{x} = P_{\mathrm{Col}\,A^\perp}(\underline{b})$

⑤ Application

- $A = [\underline{v}_1 \dots \underline{v}_d]$, $\{\underline{v}_1, \dots, \underline{v}_d\}$ basis $V = \mathrm{Col}\,A$ ⇒ $P_V(\underline{v}) = \underbrace{A(A^T A)^{-1}A^T}_{\text{coef. matrix of } P_V}\underline{v}$
- reworded:
  - $A = [\underline{v}_1 \dots \underline{v}_d]$, columns of A L.I ⇔ $A^T A$ invertible
  - Least-sq. sol:
    - $\underline{x} = (A^T A)^{-1}A^T\underline{b}$

      $A\underline{x} = \underbrace{A(A^T A)^{-1}A^T}_{\text{coef. matrix of } P_{\mathrm{Col}\,A}}\underline{b} = P_{\mathrm{Col}\,A}(\underline{b})$

12.1 Linear Transformation and Coordinates

① Recall

- $[\underline{v}]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$ ← B-coordinates of v
- $\mathcal{E}_n = \{\underline{e}_1, \dots, \underline{e}_n\}$ ⇒ $[\underline{v}]_{\mathcal{E}_n} = \underline{v}$ ← standard coordinates

② Matrix that Performs Transformation in B and C coordinates

- What if $T_A: \mathbb{R}^n \to \mathbb{R}^n$ in non-standard coordinates?
- A m×n matrix

  $B = \{\underline{b}_1, \dots, \underline{b}_n\} \subset \mathbb{R}^n$

  $C = \{\underline{c}_1, \dots, \underline{c}_m\} \subset \mathbb{R}^n$

  bases

  ⇒ $\underset{C\leftarrow B}{A} = [[A\underline{b}_1]_C \dots [A\underline{b}_n]_C]$
- Special case: $\underset{\mathcal{E}_m \leftarrow \mathcal{E}_n}{A} = A$ $\quad \underset{C\leftarrow B}{I_n} = [[\underline{b}_1]_C \dots [\underline{b}_n]_C] = \underset{C\leftarrow B}{P}$
- Formula:

  $[\underline{c}_1 \dots \underline{c}_m \mid A\underline{b}_1 \dots A\underline{b}_n]_{\text{red}} = [P_C \mid AP_B]_{\text{red}} = [I_m \mid \underset{C\leftarrow B}{A}]$
- Application:
  - $\underset{C\leftarrow B}{A}[\underline{x}]_B = [A\underline{x}]_C$
  - $\underset{C\leftarrow B}{A} = P_C^{-1} A P_B$

    $= \underset{C\leftarrow\mathcal{E}_m}{P}\ A\ \underset{\mathcal{E}_n\leftarrow B}{P}$

    standard→C; perform T_A in standard; B→standard

12.2 Simplifying Linear Transformations

① Basics

- $m \times n$ A, basis B for $\mathbb{R}^n$, basis C for $\mathbb{R}^m$
- $\underset{C\leftarrow B}{A}[\underline{x}]_B = [A\underline{x}]_C$
- $\underset{C\leftarrow B}{A} = \left[[A\underline{b}_1]_C \dots [A\underline{b}_n]_C\right] = P_C^{-1}AP = \underset{C\leftarrow \mathcal{E}_n}{P}\ A \underset{\mathcal{E}_n\leftarrow B}{P}$
- Goal: find B and C to make $\underset{C\leftarrow B}{A}$ simple

② Rank is Everything

- Rank $A = r$ $\Rightarrow$ can find bases $B = \{\underline{b}_1, \dots, \underline{b}_r, \underbrace{\underline{b}_{r+1}, \dots, \underline{b}_n}_{\text{Null } A}\} \subset \mathbb{R}^n$ $\Rightarrow$ $\underset{C\leftarrow B}{A} = \left[\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array}\right]$ ← $m\times n$

  $C = \{\underbrace{\underline{c}_1, \dots, \underline{c}_r}_{\text{Col } A}, \underline{c}_{r+1}, \dots, \underline{c}_m\} \subset \mathbb{R}^m$

  S.T. $A\underline{b}_1 = \underline{c}_1 \ \dots\ A\underline{b}_r = \underline{c}_r$

  $A\underline{b}_{r+1} = \underline{0} \ \dots\ A\underline{b}_n = \underline{0}$

- Important case: $n = m$, $B = C$ ← same coords in co- & domain
  - M and A similar $\Leftrightarrow$ $M = P^{-1}AP$ ← some invertible matrix

    $P = [\underline{b}_1 \dots \underline{b}_n]$ invertible
  - $\underset{B\leftarrow B}{A} = P_B^{-1}AP_B \Rightarrow \underset{B\leftarrow B}{A}$ similar to A

    ↑ matrices similar to A = matrices performing T_A in other linear coords

③ Simplest form for $\underset{B\leftarrow B}{A} = P_B^{-1}AP_B$

- A diagonalizable $\Leftrightarrow$ $\underset{B\leftarrow B}{A} = P_B^{-1}AP_B = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$ $\Leftrightarrow$ $A\underline{b}_1 = \lambda_1\underline{b}_1, \dots, A\underline{b}_n = \lambda\underline{b}_n$ ← A scaling B-coords directions

  ↑ Best case short of diagonal

  ↑ similar to diagonal matrix

  ↑ B eigenbasis

④ Applications

- reflections: $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ diagonalized by $\underset{B\leftarrow B}{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- rotations: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ → no b/c no non-zero vector scaled by A
- A = projection matrix $\Rightarrow$ $T_A = P_{V \parallel W}$ & $V = \text{Col } A$, $W = \text{Null } A$

  ↳ $B = \{\underbrace{\underline{b}_1, \dots, \underline{b}_r}_{\text{Col } A}, \underbrace{\underline{b}_{r+1}, \dots, \underline{b}_n}_{\text{Null } A}\}$ $\Rightarrow$ $A\underline{b}_1 = \underline{b}_1 \dots A\underline{b}_r = \underline{b}_r$, $A\underline{b}_{r+1} = \underline{0} \dots A\underline{b}_n = \underline{0}$ $\Rightarrow$ $\underset{B\leftarrow B}{A} = \left[\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array}\right]$ ← diagonal $n\times n$

  $\Rightarrow$ A projection $\Leftrightarrow$ Diagonalizable & similar $\left[\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array}\right]$

- Powers
  - $P^{-1}AP = D \Rightarrow A = PDP^{-1}$

    $\Rightarrow A^k = PD\cancel{P^{-1}P}DP^{-1} \dots PDP^{-1} = PD^kP^{-1} = P\begin{bmatrix} a_1^k & & 0 \\ & \ddots & \\ 0 & & a_n^k \end{bmatrix}P^{-1}$ (with $P^{-1}P = I_n$)

13.1 Eigenvalues, Eigenvectors, Eigenspaces

① Diagonalizable

- A diagonalizable $\Leftrightarrow$ A similar to a diagonal matrix ($P^{-1}AP = D$) $\Leftrightarrow$ There exists basis $B \subset \mathbb{R}^n$ s.t. $\underset{B\leftarrow B}{A} = \left[[A\underline{b}_1]_C \dots [A\underline{b}_n]_C\right] = P_B^{-1}AP_B$ diagonal

- A scaling in B vector direction

  $\underset{B\leftarrow B}{A} = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \Leftrightarrow A\underline{b}_1 = \lambda_1\underline{b}_1, \dots, A\underline{b}_n = \lambda_n\underline{b}_n$

- Goal: Find non-zero vectors scaled by A

② Eigenvector, Eigenvalue

- A $n\times n$ matrix $\underline{v} \in \mathbb{R}^n$ = eigenvector of A w/ eigenvalue $\lambda \in \mathbb{R}$ if:
  - $\underline{v} \neq 0$
  - $A\underline{v} = \lambda\underline{v}$
  - Note: $\underline{v} \neq \underline{0}$ but λ could be zero ← $\underline{v}$ is 0-eigenvector $\Leftrightarrow$ $\underline{v} \in \text{Null } A$

③ Eigenbasis

- A diagonalizable $\Leftrightarrow$ exists basis for $\mathbb{R}^n$ consisting of eigenvectors of A

  ↓

  $P^{-1}AP$ Diagonal $\Leftrightarrow$ columns of P = eigenvectors of A

④ Goal: Find eigenbasis

- (1) Find eigenvalues

  ↳ $\det(A - \lambda I) = 0 \Leftrightarrow \text{Null}(A - \lambda I_n) \neq 0 \Leftrightarrow (A - \lambda I_n)\underline{v} = \underline{0}$ for some $\underline{v} \neq 0$

  ↳ characteristic polynomial: $\chi_A(t) = \det(A - tI_n)$
  - λ an eigenvalue $\Leftrightarrow \chi_A(\lambda) = 0$
  - $A = \begin{bmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \Rightarrow \chi_A(t) = (\lambda_1 - t)\dots(\lambda_n - t) \Rightarrow \lambda_1 \dots \lambda_n$ are eigenvalues

- (2) Find eigenvectors

  ↳ $A\underline{v} = \lambda\underline{v} \Leftrightarrow (A - \lambda I_n)\underline{v} = \underline{0} \Leftrightarrow \underline{v} \in \text{Null}(A - \lambda I_n)$

  ↳ λ-eigenspace: subset of all λ-eigenvectors and $\underline{0}$ ← $\lambda = 0$, $E_0 = \text{Null } A$

  $E_\lambda = \text{Null}(A - \lambda I_n)$

  ↑ λ eigenvalue $\Leftrightarrow \dim E_\lambda \geq 1$

13.2 Diagonalizing Matrices

① Diagonalizing

- A diagonalizable $\Leftrightarrow P^{-1}AP = \begin{bmatrix}\lambda_1 & & 0\\ & \ddots & \\ 0 & & \lambda_n\end{bmatrix} \Leftrightarrow$ columns of P form an eigenbasis B

  $A_{B\leftarrow B}$ if $P = P_B$

  $Ab_1 = \lambda_1 b_1, \dots, Ab_n = \lambda_n b_n$

- diagonalizing A $\Leftrightarrow$ Find P w/ $P^{-1}AP$ diagonal $\Leftrightarrow$ finding eigenbasis of A

② Strategy to Diagonal Matrix A

- (1) Calculate & completely factor $\chi_A(t) = \det(A - tI_n)$
- (2) Find bases for each eigenspace $E_\lambda = \text{Null}(A - \lambda I_n)$
- (3) Take union & hope we have basis

③ Algebraic Multiplicity

- Algebraic Multiplicity: $\chi_A(t) = (\lambda_1 - t)^{m_{\lambda_1}} \cdots (\lambda_k - t)^{m_{\lambda_k}} \Rightarrow m_{\lambda_i}$
  - $m_{\lambda_1} + \dots + m_{\lambda_k} = n$ ← $\deg \chi_A(t) = n$
  - A upper triangular $\Rightarrow m_\lambda$ = number of time λ appears on main diagonal
  - $1 \le \dim E_\lambda \le m_\lambda$ ← if λ an eigenvalue, can find at most m_λ L.I λ-eigenvectors
- * unions of L.I. eigenvectors in different eigenspaces are L.I.

④ Is A Diagonalizable?

- Criterion: A diagonalizable $\Leftrightarrow \dim E_\lambda = m_\lambda$
- A has n distinct eigenvalues $\Rightarrow$ A diagonalizable

  $m_\lambda = 1$ and $1 \le \dim E_\lambda \le m_\lambda = 1 \Rightarrow \dim E_\lambda = m_\lambda = 1$

- not distinct eigenvalues

  ↳ Alg mult $m_\lambda > 1$ → need to calculate $\dim E_\lambda = \text{Null}(A - \lambda I)$

  → if $\dim E_\lambda < m_\lambda$

  ⇒ not diagonalizable

  → # free columns = # dim

14.1 Symmetric Matrices and the Spectral Theorem

① Recall Diagonalizable

- A diagonalizable $\Leftrightarrow P^{-1}AP = \begin{bmatrix}\lambda_1 & & 0\\ & \ddots & \\ 0 & & \lambda_n\end{bmatrix} \Leftrightarrow$ columns of P form an eigenbasis B
- Diagonalizing A = find P w/ $P^{-1}AP$ diagonal = Find eigenbasis of A

⇒ study diagonalization for symmetric matrices

② Symmetric Matrix

- A symmetric $\Leftrightarrow A^T = A$
- Special properties
  - $u \cdot (Av) = (A^T u) \cdot v \Rightarrow$ if A symmetric: $u \cdot (Av) = (Au) \cdot v$
- A symmetric ⇒ All eigenvalues are real
- A symmetric ⇒ A preserves V ⇒ A preserves $V^\perp$

  ⇒ A preserves $V \neq \{0\}$ ⇒ V contains an eigenvector of A

- Note: A orthogonally diagonalizable $\Leftrightarrow P^T AP = \begin{bmatrix}\lambda_1 & & 0\\ & \ddots & \\ 0 & & \lambda_n\end{bmatrix} \Leftrightarrow$ columns of P form an orthogonal eigenbasis

  diagonal matrices ortho. diagonalized by standard basis

③ Spectral Theorem

- A real symmetric ⇔ A orthogonally diagonalizable
  - $\{b_1, \dots, b_d\}$ eigenvectors of A ⇓ A preserves V $\xrightarrow{\text{A symmetric}}$ A preserves $V^\perp$ ⇑ $V^\perp$ contains an eigenvector b_{d+1} of A
  - * $d = n$ & $\{b_1, \dots, b_n\}$ is orthonormal eigenbasis
- A symmetric ⇒ Different eigenspaces are orthogonal

④ Orthogonally Diagonalize

- compute $\chi_A(t)$ & factor
- for each λ s.t. $\chi_A(\lambda) = 0$, compute basis for $E_\lambda = \text{Null}(A - \lambda I_n)$
- for any i s.t. $\dim E_{\lambda_i} > 1$, run G.S. of basis E_{λ_i} to get orthogonal basis
- normalize
- take union
- $P^T AP$ → orthogonally diagonalize any real symmetric matrix

14.2 Singular Value Decomposition

- Present linear transformation $T_A: \mathbb{R}^n \to \mathbb{R}^m$ in simplest way

① Recall Rank is Everything

- given A $m\times n$ matrix, there exists orthonormal bases:

$B = \{\underline{b}_1, \dots, \underline{b}_n\} \subset \mathbb{R}^n$ and $C = \{\underline{c}_1, \dots, \underline{c}_m\} \subset \mathbb{R}^m$ s.t.

$A\underline{b}_1 = \underline{c}_1, \dots, A\underline{b}_r = \underline{c}_r, \; A\underline{b}_{r+1} = \underline{0}, \dots A\underline{b}_n = \underline{0}$

$\Rightarrow \underset{C\leftarrow B}{A} = P_C^{-1} A P_B = \left[\begin{array}{c:c} \begin{smallmatrix}1 & & 0\\ & \ddots & \\ 0 & & 1\end{smallmatrix} & 0 \\ \hdashline 0 & 0\end{array}\right]$ rank $= r$

② Singular Value Decomposition

- restrict ourselves to orthonormal bases → orthonormally diagonalize
  - ↳ eigen as V
  - ↳ U w/ SVD

BVT
↓
SVD

- given A $m\times n$ matrix, there exists orthonormal bases

$V = \{\underline{v}_1, \dots, \underline{v}_n\} \subset \mathbb{R}^n$ and $U = \{\underline{u}_1, \dots, \underline{u}_m\} \subset \mathbb{R}^m$ s.t.

$A\underline{v}_1 = \sigma_1 \underline{u}_1, \dots, A\underline{v}_r = \sigma_r \underline{u}_r, \; A\underline{v}_{r+1} = \underline{0}, \dots A\underline{v}_n = \underline{0}$

$\Rightarrow \underset{U\leftarrow V}{A} = U^T A V = \Sigma = \left[\begin{array}{c:c} \begin{smallmatrix}\sigma_1 & & 0\\ & \ddots & \\ 0 & & \sigma_r\end{smallmatrix} & 0 \\ \hdashline 0 & 0\end{array}\right]$ rank $= r$, $\sigma_1 \geq \sigma_2 \dots \geq \sigma_r > 0$ ↳ singular values

$U = [\underline{u}_1 \dots \underline{u}_m]$ $V = [\underline{v}_1 \dots \underline{v}_n]$
orthogonal matrices

$\Rightarrow A = U \Sigma V^T$ ← SVD of A

- A^TA orthogonally diagonalized by V
- Spectral Theorem ⇒ SVD ← All matrices admit a SVD

↑ symmetric matricies are orthogonally diagonalizable

③ SVD Algorithm

- orthogonally diagonalize A^TA
- reorder so eigenvalues decreasing $\underset{\underline{v}_1}{\lambda_1} \geq \underset{\underline{v}_2}{\lambda_2} \geq \dots \geq \underset{\underline{v}_n}{\lambda_n} \geq 0$
  - ↳ define $V = [\underline{v}_1 \dots \underline{v}_n]$
- $\sigma_i = \sqrt{\lambda_i}$. we can define Σ

define $\underline{u}_1 = \frac{1}{\sigma_1} A\underline{v}_1, \dots, \underline{u}_r = \frac{1}{\sigma_r} A\underline{v}_r \Rightarrow \{\underline{u}_1, \dots, \underline{u}_r\}$ ONB for Col A

- extend $\{\underline{u}_1, \dots, \underline{u}_r\}$ to ONB for $\mathbb{R}^m$
  - ↳ find a basis for $(\text{Col } A)^\perp = \text{null } A^T$
  - ↳ Apply G.S. to get orthogonal basis for $(\text{Col } A)^\perp$
  - ↳ normalize to get $\{\underline{u}_{r+1}, \dots, \underline{u}_m\}$
- define U & write SVD → $A = U^T \Sigma V$ $A = U\Sigma V^T$